

Multi-Source Candidate Data Transformer — Technical Design

Eightfold Engineering Intern Assignment (Jul-Dec 2026)

1. Pipeline Breakdown

The pipeline is a linear sequence of well-isolated stages. Each stage reads its predecessor's output; no stage reaches back or mutates an earlier one.

Detect — File extension → source type. Unrecognised or missing files are skipped (stderr warning). Never crashes.

Extract — One extractor per source type. Returns a list of RawRecords, each field wrapped in a ProvenancedValue(value, source, method, confidence). Errors in one record do not abort the batch.

Normalize — Deterministic, dependency-free. Phones → E.164 (best-effort; left null if digits can't be reconciled). Dates → YYYY-MM (or sentinel "present"). Skills → canonical alias map (js→JavaScript, k8s→Kubernetes, ...). Country → ISO-3166 alpha-2, or null if unknown.

Match / Group — Union-find over shared keys: exact normalized email (strongest) > exact phone > exact lowercased full name. Transitive grouping: if CSV shares email with ATS and ATS shares phone with resume, all three collapse to one candidate.

Merge — Scalar fields: highest (source_trust × extraction_confidence) wins. Trust order: ATS JSON ≥ CSV > resume > notes. List fields (emails, phones, skills): union, de-duped by normalized key, confidence boosted +0.1 per extra corroborating source.

Confidence — overall_confidence = mean of per-field (trust × confidence) scores across all populated fields. Every field in provenance[] records {field, source, method}.

Project — Config-driven projection onto the requested output shape. Supports: field subset, rename/remap (e.g. emails[0] → primary_email, skills[].name → flat list), per-field normalize hints, include_confidence / include_provenance toggles, on_missing: null | omit | error. The canonical record is never mutated.

Validate — Checks projected output against the type annotations in the same config (string, number, object, string[], object[]). Errors are collected and returned alongside the output, not thrown silently.

2. Canonical Output Schema & Normalization Choices

The internal canonical record (before projection) always contains all fields:

Field	Type	Normalization
candidate_id	string	SHA-1 of match key, hex prefix
full_name	string null	title-cased from highest-trust source
emails	string[]	lowercased, RFC-5322 pattern
phones	string[]	E.164 (+CC digits); null if ambiguous
location	{city,region,country}	country → ISO-3166 alpha-2 or null
links	{linkedin,github,portfolio,other[]}	https:// prefix enforced
headline	string null	raw text, highest-trust source
years_experience	number null	numeric, no rounding
skills	[[name,confidence,sources[]]]	canonical alias map; cross-source boost
experience	[[company,title,start,end,summary]]	dates as YYYY-MM or "present"
education	[[institution,degree,field,end_year]]	end_year int or null
provenance	[[field,source,method]]	every populated field traced
overall_confidence	number 0..1	weighted mean of field scores

3. Merge & Conflict-Resolution Policy

- **Match keys:** normalized email (highest trust) → normalized phone → exact lower-cased full name. Union-find ensures transitive grouping with no duplicate profiles.
- **Scalar winner:** field value = $\text{argmax}(\text{source_trust} \times \text{extraction_confidence})$. Structured sources dominate free text; a CSV email (0.95) beats a resume regex email ($0.8 \times 0.75 = 0.60$).
- **List union:** emails, phones, skills are merged across all sources, de-duplicated on their normalized key. Each surviving entry keeps the best-scoring ProvenancedValue.
- **Skill corroboration:** confidence ± 0.1 per additional source that mentions the same skill (capped at 0.99). A skill seen in both ATS JSON and a resume is more trustworthy than one only in a recruiter note.
- **Experience / education:** unioned and de-duped by (company, title) or (institution). More detailed record (has summary, has dates) preferred.

4. Runtime Config / Projection

Clean separation between the canonical record (always fully populated) and the projected output (shaped per config). The config can:

- Select a subset of fields via path.
- Remap from a canonical path: "from": "emails[0]" or "skills[].name".
- Set per-field normalization hints ("normalize": "E164").
- Toggle provenance and confidence blocks.
- Choose on_missing: null | omit | error per run.

Projection is validated against the types declared in the same config before the result is returned.

5. Edge Cases & How We Handle Them

- **Malformed/missing source file:** extractor returns [] and logs to stderr. Pipeline continues with remaining sources.
- **Blank CSV rows:** skipped (no name, email, or phone present) rather than emitted as a phantom candidate.
- **Garbage JSON:** json.JSONDecodeError caught; returns []. Wrong-but-confident is worse than honestly empty.
- **Ambiguous phone:** if we cannot determine E.164 unambiguously we return null rather than invent digits. Caller sees a missing-field warning.
- **Unknown country:** left null rather than guessing an ISO code.
- **Same email, different casing:** all emails normalized to lowercase before de-dup, so "Priya@gmail.com" and "priya@gmail.com" collapse to one entry.

6. What We Would Deliberately Leave Out Under Time Pressure

- Live LinkedIn/GitHub API fetching (auth complexity, rate limits; links extracted passively from resumes instead).
- Fuzzy / ML-based name matching (deterministic exact match is explainable and safe for a first version; fuzzy matching risks silent merges of different people).
- Layout-aware resume parser (section detection is heuristic/regex; a production system would use a dedicated parser or LLM extraction step).
- Full skills taxonomy (alias map demonstrates the mechanism; real deployment would back it with a maintained ontology).